THE HOM FUNCTOR FROM A FINITE LOCALLY FREE SOURCE TO A SEPARATED LOCALLY QUASI-FINITE TARGET IS REPRESENTABLE BY A SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 20256-20262 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20264-20268, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Both cited lemmas are already on record (S9 and this lane); no new author-side verification performed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma hom-from-finite-locally-free-representable (S9 registered); chapter lemma separated-locally-quasi-finite-over-affine (this lane).

Source reference. Original lemma 0BL4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20201–20209. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07S0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12662–12666. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $Z \to S$ and $X \to S$ be morphisms of schemes. Assume $Z \to S$ is finite locally free and $X \to S$ is separated and locally quasi-finite. Then $Mor_S(Z, X)$ is representable by a scheme.*

Proof. This follows from Lemmas 0BL4 and 07S0. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.